

# AAPL Stock Market Analysis Insight Summary Report

AnalystLab Africa | Week 6 | Batch A

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## 1. Dataset Overview

The dataset used for this analysis is the Apple Inc. (AAPL) historical stock market data sourced from Yahoo Finance via the yfinance Python library. It covers a 5-year period from June 21, 2021 to June 18, 2026, capturing 1,255 trading days of real market activity.

| Attribute          | Details                              |
|--------------------|--------------------------------------|
| Source             | Yahoo Finance (via yfinance)         |
| Ticker             | AAPL (Apple Inc.)                    |
| Date Range         | June 21, 2021 – June 18, 2026        |
| Total Trading Days | 1,255 rows                           |
| Features           | Date, Open, High, Low, Close, Volume |
| Missing Values     | None                                 |
| Duplicates         | None                                 |

## 2. Data Transformation Techniques Applied

The following transformation and feature engineering techniques were applied:

**Data Cleaning:** Verified zero missing values, duplicates, and invalid price entries (High < Low). Data was sorted chronologically and date column confirmed as datetime format.

**Time Feature Extraction:** Extracted Year, Month, Month Name, Quarter, and Day of Week from the Date column to enable time-based grouping and trend analysis.

**Daily Price Change:** Calculated as Close minus Open price, giving the absolute and percentage price movement per trading day.

**Daily Return:** Computed as the percentage change in closing price from one day to the next using `pct_change()` — the primary metric for return analysis.

**Moving Averages:** Created 7-day, 30-day, and 90-day rolling averages of the closing price to smooth short-term noise and reveal underlying trends.

**Volatility Measure:** 30-day rolling standard deviation of daily returns — used as the primary risk indicator for the investor.

**Monthly Returns:** Grouped by year and month to calculate the percentage change from the first to last closing price of each month.

**Price Range:** Computed as High minus Low per day — indicating intraday price movement width.

## 3. Trends Identified

**Long-Term Uptrend:** AAPL closed at \$128.96 in June 2021 and reached \$298.01 by June 2026 — a 131.1% total return over 5 years. The overall price direction is strongly bullish, rewarding patient investors.

**2022 Bear Market:** The stock experienced a significant decline through 2022, with the trough price reaching \$122.93. This was the highest-risk period, with 30-day volatility peaking above 3%. Investors who sold during this period locked in losses.

**Strong Recovery (2023–2026):** From early 2023, AAPL entered a sustained recovery, driven by consistent upward momentum. By 2025, the price had surpassed its previous highs, reaching a peak of \$315.20.

**April 2025 Extreme Volatility:** The single best day (+15.33% on April 9, 2025) and worst day (-9.25% on April 3, 2025) both occurred within the same week in April 2025 — indicating a major market event that caused extreme short-term swings before recovery.

**Volume Spikes Signal Risk:** Trading volume spiked significantly above the 65M daily average during periods of high uncertainty. These spikes often coincided with large price movements, making volume a reliable early warning signal for investors.

**Seasonal Patterns:** The monthly returns heatmap reveals that September consistently delivers negative average daily returns across multiple years, while May and July tend to be stronger months for AAPL performance.

#### 4. Features Created

| Feature                | Description                 | Purpose                    |
|------------------------|-----------------------------|----------------------------|
| Daily_Price_Change     | Close - Open                | Intraday direction         |
| Daily_Price_Change_Pct | % change Open to Close      | Relative intraday movement |
| Price_Range            | High - Low                  | Intraday volatility width  |
| Daily_Return           | % change Close to Close     | Core return metric         |
| MA_7                   | 7-day rolling avg of Close  | Short-term trend           |
| MA_30                  | 30-day rolling avg of Close | Medium-term trend          |
| MA_90                  | 90-day rolling avg of Close | Long-term trend            |
| Volatility_30          | 30-day std of Daily Return  | Risk indicator             |
| Monthly_Return         | Month start to end % change | Monthly performance        |
| Year / Month / Quarter | Extracted from Date         | Time-based grouping        |

#### 5. Key Insights from the Analysis

| Metric                 | Value    |
|------------------------|----------|
| Start Price (Jun 2021) | \$128.96 |
| End Price (Jun 2026)   | \$298.01 |
| 5-Year Total Return    | 131.1%   |
| Peak Price             | \$315.20 |
| Trough Price           | \$122.93 |

|                                 |                         |
|---------------------------------|-------------------------|
| <b>Best Single Day</b>          | +15.33% (April 9, 2025) |
| <b>Worst Single Day</b>         | -9.25% (April 3, 2025)  |
| <b>Avg 30-Day Volatility</b>    | 1.63%                   |
| <b>Avg Daily Trading Volume</b> | 65 Million shares       |

1. AAPL more than doubled in value over 5 years — an investor who held from June 2021 to June 2026 earned a 131.1% return.
2. The 2022 bear market was the greatest test of investor patience. Volatility peaked above 3% and the price fell to its lowest point of \$122.93. However, the stock fully recovered and went on to reach \$315.20.
3. The April 2025 volatility event shows that even mature stocks like AAPL are vulnerable to sudden market shocks — but short-term pain did not derail the long-term trend.
4. Moving average crossovers were reliable momentum signals. When the 7-day MA crossed above the 30-day MA, upward price movement followed in most cases.
5. September is historically the weakest month for AAPL returns — investors should exercise caution and monitor positions more closely during this period.
6. High trading volume days (above 65M shares) consistently coincided with significant price movements — making volume a practical early warning tool for risk management.

## 6. Recommendations

**Hold During Volatility Spikes:** The data shows that AAPL consistently recovered after periods of high volatility. Selling during panic periods (like 2022) would have resulted in significant losses. A long-term hold strategy was the optimal approach.

**Use Moving Averages as Entry/Exit Signals:** Monitor the 7-day and 30-day moving averages. When the 7-day MA crosses above the 30-day MA, it signals building momentum — a potential entry point. When it crosses below, it signals caution.

**Watch Volume as a Risk Indicator:** Any trading day where volume significantly exceeds the 65M daily average should be treated as a risk alert. High volume often precedes large price swings in either direction.

**Reduce Exposure in September:** Historical data consistently shows September as the weakest month for AAPL returns. Investors may consider reducing position size or setting tighter stop-losses during this period.

**Use the 90-Day MA as a Long-Term Support Level:** The 90-day moving average acted as a reliable support level throughout the 5-year period. Price drops toward the 90-day MA historically presented strong buying opportunities.